

Nathaniel Fink-Humes

nfinkhum@gmu.com

EDUCATION

GEORGE MASON UNIVERSITY

Mathematics Ph.D, in Progress

Advisor: Carlos Rautenberg

Attending Fall 2024 – Present

UNIVERSITY OF VIRGINIA

Mathematics Bachelor of Arts Cum Laude

GPA 3.916

Attended Fall 2022 – Spring 2024

NORTHERN VIRGINIA COMMUNITY COLLEGE

Mathematics Associate of Sciences Summa Cum Laude

GPA 4.0

Attended Fall 2019 – Spring 2022

RESEARCH INTERESTS

Analysis of Partial Differential Equations, Variational Methods, Variational Inequalities, Quasi-Variational Inequalities, Mathematics of Thermistors and Thermoforming.

PRESENTATIONS

Fink-Humes, N. & Rautenberg C., “Analysis of a two membrane thermoelastic problem”, *Seminar Talk, Student Research Talks (StReeTs)*, George Mason University. Fairfax, VA. February 2026

Fink-Humes, N. & Rautenberg C., “Analysis of a two membrane thermoelastic problem”, *Seminar Talk, Analysis Seminar*, George Mason University. Fairfax, VA. February 2026

Fink-Humes, N. & Rautenberg C., “Analysis of a quasi-variational two membrane problem arising in thermoelasticity.” *Oral Presentation, Joint Mathematical Meetings AMS Contributed Paper Session on Mathematical Physics I*, Washington, DC. January 2026

RELEVANT WORK EXPERIENCE

UNIVERSITY OF VIRGINIA, *Math Grader and Tutor*, VA

08/2023 — 05/2024

- o Tutor students in Differential Equations and Real Analysis.
- o Grade quizzes, exams, and homework for 140 students across 3 sections of UVA’s Financial Mathematics course.
- o Grade 24 students in UVA’s proof-based Advanced Linear Algebra course.

DEVILS BACKBONE BREWING COMPANY, *Springfield Brewery Representative*, VA

08/2018 — 12/2019

- o Independently manage work hours, location, and strategies towards accomplishing long term goals.
- o Develop and implement training programs for utilizing mathematics to drive sales conversations.
- o Drive 23.1% increase in volume over my tenure.
- o Plan and execute innovative events to engage consumers with new and old products.
- o Train accounts and wholesalers on the process of brewing, beer styles, beer history, etc.
- o Redesign forecasting program for product allocations across multiple wholesaler warehouses.

FORDHAM AND DOMINION BREWING COMPANY, *Virginia Market Manager*, VA

02/2018 — 08/2018

- o Reversed an over yearlong 30% month-to-month downward trend in sales.
- o Built my own CRM using Microsoft Access to track various individual account data.
- o Repair damaged relationships with wholesalers and accounts.
- o Cold call and activate new accounts.

OTHER WORK EXPERIENCE

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DOMINION WINE & BEER, *Retail Sales Associate*, VA 01/2020 — 05/2022

- o Utilize detailed knowledge of beer to match products with customers' wants towards developing return customers.
- o Open and close the business including managing drawers and deposits.
- o Function as Manger on Duty while working for the purposes of dealing with customers or vendors.

ARROWINE & CHEESE, *Assistant Manager and Beer Buyer*, VA 02/2015 — 02/2018

- o Cultivate relationships with customers, make product recommendations, and anticipate future wants.
- o Write and publish weekly newsletters to 1900+ contacts with an open-rate 15% above industry average.
- o Develop comprehensive knowledge of all beers available in the area and curate highly sought selection.

ARROWINE & CHEESE, *Cheesemonger*, VA 10/2013 — 02/2015

- o Utilized comprehensive knowledge of all available cheeses (~180-230) to accommodate customer needs.
- o Assisted Beer and Wine Departments with stocking and sales.

GREAT HARVEST BREAD COMPANY(defunct), *Farmer's Market Manager*, VA 08/2011 — 06/2012

- o Oversaw planning, on-site setups, product sales, and marketing for all 4 local Farmer's Markets.
- o Redesigned the store's Sandwich Program to increase customer options and reduce food waste.

CERTIFICATIONS

BEER JUDGE CERTIFICATION PROGRAM 08/2019

Recognized Judge

ACADEMIC AWARDS AND SCHOLARSHIPS

GEORGE MASON UNIVERSITY

Presidential Scholarship 08/2024-Present

UNIVERSITY OF VIRGINIA

Cum Laude 05/2024

NORTHERN VIRGINIA COMMUNITY COLLEGE

Summa Cum Laude 05/2022

Presidential Scholar 05/2022

Presidential Scholar 12/2021

Presidential Scholar 05/2020